

# Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

**Benchmark:** MENTALCLOUDS | **Context:** Global South Mental Health Practitioners — India-Anchored OMHC / Peer-Support Deployment

**Input Content definition:** Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

The benchmark's authors explicitly state that 'the counseling sessions in this work represented a certain demographic region (American) and thus may not apply to therapy counseling for other demographics' **[Q70]**, with sessions sourced from YouTube **[Q12, Q60]**. The deployment population is mixed — India-anchored with a meaningful share of non-Indian Global South contexts (deployment\_context, A1). Although the elicitation notes that CBT protocols 'transfer reasonably' across regions (A1), critical India-specific stressors (caste, dowry, somatization, faith-healer integration **[WEB-15]**) and Sub-Saharan African conceptualizations (spiritual forces, ancestral relationships **[WEB-16]**) are structurally absent from American session content. The Lancet Digital Health viewpoint **[WEB-29]** explicitly identifies cultural and linguistic characteristics of mental-health expression as a key barrier for English-Western-trained LLMs. No compensating Global South counseling-summarization benchmark exists **[WEB-27, WEB-28]**, so transfer cannot be triangulated. The claim of 'heterogeneous demographic spectrum' **[Q13]** refers to within-American diversity, not cross-cultural diversity. Additionally, the YouTube provenance raises ecological-validity concerns: 'natural conversations are incoherent and grammatically unfluent' **[Q61]** and likely reflect mock or semi-public exchanges, which may differ from real OMHC clinical transcripts in disclosure depth and risk-content density.

| ID            | Evidence                                                                                                                                                                                                                                                                                                                          |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q12</b>    | "Comprising 11.5K utterances extracted from 191 counseling sessions involving therapists and patients, this dataset draws from publicly accessible platforms such as YouTube." (p.6)                                                                                                                                              |
| <b>Q13</b>    | "Embracing a heterogeneous demographic spectrum with distinctive mental health concerns and diverse therapists, the dataset facilitates the formulation of a comprehensive and inclusive approach for researchers." (p.6)                                                                                                         |
| <b>Q60</b>    | "The counseling dataset was curated from multiple multimedia online sources such as youtube transcripts [37]." (p.15)                                                                                                                                                                                                             |
| <b>Q61</b>    | "Hence, most of these natural conversations are incoherent and grammatically unfluent." (p.15)                                                                                                                                                                                                                                    |
| <b>Q70</b>    | "Additionally, the counseling sessions in this work represented a certain demographic region (American) and thus may not apply to therapy counseling for other demographics." (p.16)                                                                                                                                              |
| <b>WEB-15</b> | Faith-healer integration documented in Indian mental health practice ( <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC7616029/">https://pmc.ncbi.nlm.nih.gov/articles/PMC7616029/</a> )                                                                                                                                        |
| <b>WEB-16</b> | Sub-Saharan African mental health frequently attributed to spiritual forces, ancestors, witchcraft ( <a href="https://cdn.techscience.cn/files/jpa/2025/TSP_JPA-35-1/TSP_JPA_65764/TSP_JPA_65764.pdf">https://cdn.techscience.cn/files/jpa/2025/TSP_JPA-35-1/TSP_JPA_65764/TSP_JPA_65764.pdf</a> )                                |
| <b>WEB-27</b> | Africa Health Check covers medical LLM bias but not mental health summarization ( <a href="https://aclanthology.org/2025.emnlp-main.1639.pdf">https://aclanthology.org/2025.emnlp-main.1639.pdf</a> )                                                                                                                             |
| <b>WEB-28</b> | JMIR systematic review: emerging Global South medical benchmarks (AfriMed-QA, TRINDs) but limited representation ( <a href="https://www.jmir.org/2025/1/e84120/PDF">https://www.jmir.org/2025/1/e84120/PDF</a> )                                                                                                                  |
| <b>WEB-29</b> | Lancet Digital Health 2025: cultural/linguistic characteristics pose major challenges for English/Western-trained LLMs in mental health ( <a href="https://www.thelancet.com/journals/landig/article/PIIS2589-7500(24)00255-3/fulltext">https://www.thelancet.com/journals/landig/article/PIIS2589-7500(24)00255-3/fulltext</a> ) |

## Strengths

- The benchmark authors explicitly disclose the American demographic limitation **[Q70]**, enabling downstream users to make informed decisions about transfer.
- The English-only standard register matches the deployment's stated dataset (A4), so for the subset of Indian deployment data resembling broadly standard English CBT exchanges, content overlap exists **[Q12, Q60]**.

- The dataset embraces some demographic diversity within its source context [Q13], providing within-source variation across mental health concerns and therapists.

| ID  | Evidence                                                                                                                                                                                                                  |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q12 | “Comprising 11.5K utterances extracted from 191 counseling sessions involving therapists and patients, this dataset draws from publicly accessible platforms such as YouTube.” (p.6)                                      |
| Q13 | “Embracing a heterogeneous demographic spectrum with distinctive mental health concerns and diverse therapists, the dataset facilitates the formulation of a comprehensive and inclusive approach for researchers.” (p.6) |
| Q60 | “The counseling dataset was curated from multiple multimedia online sources such as youtube transcripts [37].” (p.15)                                                                                                     |
| Q70 | “Additionally, the counseling sessions in this work represented a certain demographic region (American) and thus may not apply to therapy counseling for other demographics.” (p.16)                                      |

## Checklist

### Checklist item definitions:

| ID   | Assessment Question                                                        |
|------|----------------------------------------------------------------------------|
| IC-1 | Do inputs require region-specific cultural/geographic/dialectal knowledge? |
| IC-2 | Does culturally sensitive content align with target culture?               |
| IC-3 | Flag inputs assuming Western-specific knowledge                            |
| IC-4 | Regional annotator recruitment for culturally sensitive instances          |
| IC-5 | Document content issues harming content validity                           |

### Checklist responses:

**IC-1:** Yes — the deployment requires region-specific knowledge (Indian family-system framing, somatization patterns, caste/dowry-related stressors, faith-healer integration [WEB-15]; Sub-Saharan African spiritual/ancestral conceptualizations [WEB-16]) that are not present in American YouTube-sourced sessions [Q70].

**IC-2:** Partial misalignment: CBT-anchored disclosure overlaps (A1, A3), but India-specific and broader Global South cultural disclosure patterns are structurally absent from American session content [Q70, WEB-29].

**IC-3:** American-specific knowledge in the sessions (e.g., American healthcare-system references, US-specific stressors) may introduce construct-irrelevant variance for Global South practitioners — though specific instances could not be verified without dataset inspection.

**IC-4:** INSUFFICIENT DOCUMENTATION — the paper does not document recruitment of regional (Indian or other Global South) annotators to flag culturally sensitive instances at the input-content level; reference annotation creator credentials are also undocumented (annotator\_representativeness.reference\_annotation\_creators flagged as needing verification).

**IC-5:** Documented harms to content validity: explicit American demographic bias [Q70]; YouTube-sourced sessions characterized as 'incoherent and grammatically unfluent' [Q61], suggesting mock/semi-public rather than naturalistic clinical material; absence of South Asian and broader Global South cultural framings; no compensating Global South benchmark exists for triangulation [WEB-27, WEB-28].

| ID     | Evidence                                                                                                                                                                                                                                                                                           |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q61    | “Hence, most of these natural conversations are incoherent and grammatically unfluent.” (p.15)                                                                                                                                                                                                     |
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### Information Gaps

- Exact proportional split of India vs. other Global South content in deployment data (deferred in regional YAML).
- MEMO dataset annotator credentials and inter-annotator agreement at reference-summary creation stage.

### Requires Expert Verification

- Whether YouTube-sourced sessions are mock/scripted vs. naturalistic — needs direct inspection of dataset samples.
- Whether American YouTube content contains incidental cross-cultural representation that partially mitigates the demographic limitation.

### Remediation

**Gap 1: Sessions are American-sourced YouTube transcripts; Indian and broader Global South cultural disclosure patterns are structurally absent.**

**Recommendation:** Curate a supplementary mixed-cultural session corpus drawn from Indian OMHC partners (under appropriate consent and DPDP Act compliance) and at least one Sub-Saharan African or Southeast Asian counseling source, and treat this as a separate evaluation slice rather than a replacement.

**Gap 2: Ecological-validity concern: YouTube-sourced sessions described as 'incoherent and grammatically unfluent' [Q61] and likely mock/semi-public, possibly differing from naturalistic OMHC sessions.**

**Recommendation:** Conduct a side-by-side comparison study on a small sample of real OMHC transcripts (under IRB/consent) vs. benchmark sessions to quantify ecological-validity drift before relying on benchmark scores for deployment decisions.

| ID  | Evidence                                                                                       |
|-----|------------------------------------------------------------------------------------------------|
| Q61 | "Hence, most of these natural conversations are incoherent and grammatically unfluent." (p.15) |